

Item 1 - Take-home Task

Jump To

1. Choice of Structure and Representation	2
Task	2
Questions	2
Submit	2
2. Decomposition and Function Design	3
Task	3
Submit	3
3. Iteration Choices and Trade-offs	4
Task	4
Questions	4
Submit	4

1. Choice of Structure and Representation

Task:

You will analyse a small collection of plain text log lines representing simple system messages. You must solve the same analysis problem in exactly two different ways, each using a different primary data structure. The goal is not to produce the shortest code, but to demonstrate informed choice of data representation and to reason about its consequences.

Questions:

1. For each solution, explain why the chosen data structure is a good fit for this problem.
2. For each solution, explain what tasks become easier and what tasks become harder as a result of using this data structure.
3. For each solution, identify at least one error that an inexperienced programmer is likely to make when using this data structure in this context.

Submit:

- Exactly two short Python scripts which analyse the same set of log lines using different primary data structures (for example, one list based solution and one set or dictionary based solution).
- A short written explanation document (maximum 300 words total) addressing the questions listed below for both solutions.

2. Decomposition and Function Design

Task:

You will design a small function based solution to a multi step text processing problem. The problem should be solved by decomposing the work into simple, reusable functions with clear responsibilities. The focus is on structure and clarity rather than size.

Questions:

None.

Submit:

- One Python script implementing at least three functions with clear and distinct responsibilities.
- Docstrings for each function describing inputs, outputs, and assumptions.
- At least three demonstrative test calls per function, including example input and expected output, written as comments or simple print statements.

3. Iteration Choices and Trade-offs

Task:

You will process a large plain text data file containing many lines of simple numeric values. You must solve the same processing task in three different ways, using a for loop, a list comprehension, and a generator. The goal is to compare iteration styles and explain their trade-offs.

Questions:

1. Explain when each iteration approach is appropriate.
2. Explain what happens during evaluation for each approach.
3. State which approach you would choose under time pressure and justify your choice.

Submit:

- Three short Python solutions implementing the same processing task using a for loop, a list comprehension, and a generator respectively.
- A short written explanation document (maximum 250 words) answering the questions listed below.